Quantiles

: Statistical measure used to divide a set of numerical Data into equal size of groups, with each group containing an equal no. of observation.

① **Quartile** :- We Divide the data into four equal part

ex :- 100% data (4 part will be -
$Q_1 = 25\%$, $Q_2 = 50\%$, $Q_3 = 75\%$, $Q_4 = 100$

② **Quintile** :- Divide the data into 5 equal parts.

ex :-

20%	20%	20%	20%	20%

\- 100%

③ **Deciles** :- Divide the data into 10 equal parts.

ex :-

10%	10%	10%	-	-	.	.	-	10%

\- 100%

④ **Percentile** :- Divide the data into 100 equal parts.

$$PL = \frac{P}{100}(N+1)$$

P → Percentile

N → Total no. of observations

PL → Percentile Location

Note :- PL = It is the Location not the exact no. in the Dataset.

:- After sorting the Data.

ex :-

data = [78, 82, 84, 88, 91, 93, 94, 96, 98, 99] (N = 10)

(positions 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

What is the percentile of 75% ?

$$PL = \frac{P}{100}(N+1) = \frac{\cancel{75}^{3}}{\cancel{100}_{4}}(10+1) = \frac{3}{4}(11)$$

$$= \frac{33}{4} = \boxed{8.25}$$

$\boxed{PL = 8.25}$ → Location

$8^{th} = 96 \quad 9^{th} = 98$

$= 8^{th} + 0.25(9^{th} - 8^{th})$

$= 96 + 0.25(98 - 96)$

$= 96 + 0.25(2)$

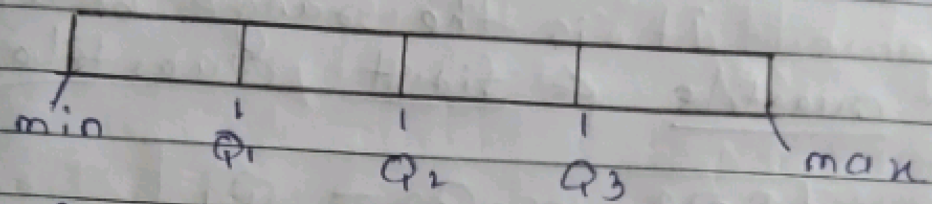
$\boxed{= 96.5}$ → 75^{th} percentile

Date.

Outlier Detection → Box plot

↳ 5 Number summary

The five no. summary is a descriptive statistics that provides a summary of a dataset. It consist of five values that divide the data into four equal parts (also know as outlier)



① min (Lower-boundary) :- It is the smallest possible value in the dataset.

→ The value may or may not be available in the dataset.

$$\boxed{LB = Q_1 - 1.5\,(IQR)}$$

② Q_1 = It is the first Qurtile i.e. 25% of the data

$$\frac{P}{100}(N+1)$$

③ Q_2 = It is the second Quartile i.e. 50% of the data.

④ Q_3 = It is third Quartile i.e. 75% of the data.

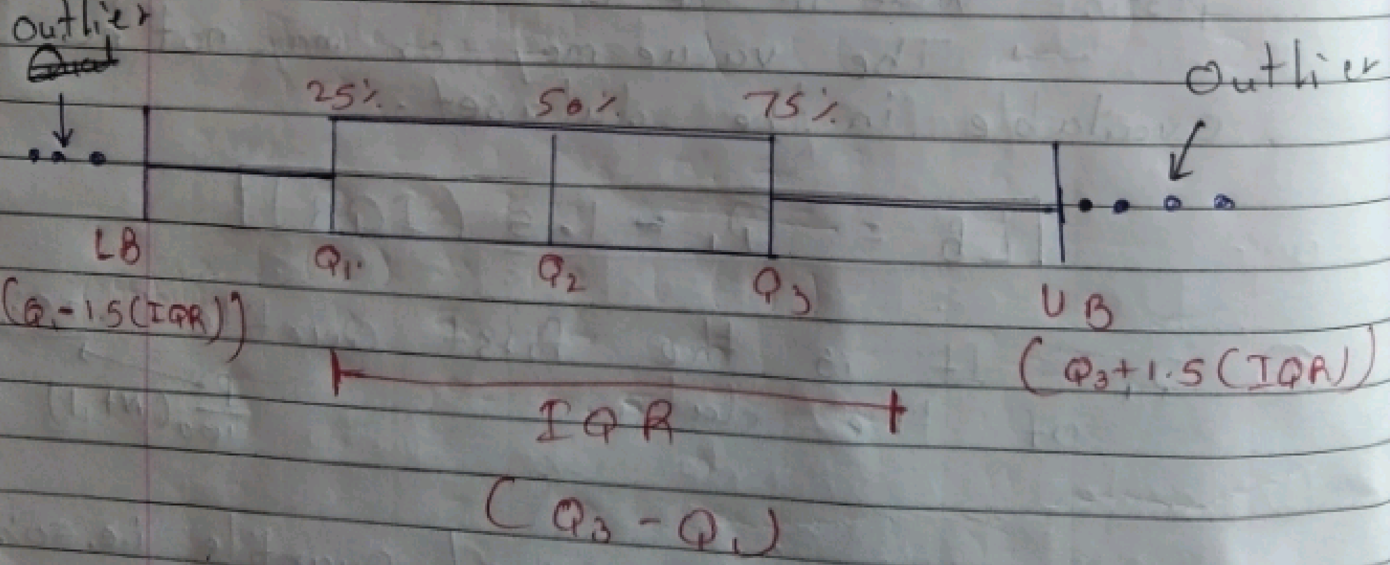
(5) Max → Upper-boundary :- It is the possible max value in the dataset

→ It may / may not be available in the dataset.

$$\boxed{Max = Q_3 + 1.5 (IQR)}$$

(6) IQR :- Inter Quartile Range :- It is the range between third quartile & first Quartile

$$\boxed{IQR = Q_3 - Q_1}$$



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Eng

6, 213, 241, 260, 281 $\uparrow^{290}$ 314, 321, 350, 1560

1 2 3 4 5 6 7 8 9 10

$$p^L = \frac{P}{100}(N+1)$$

$Q_1 = \frac{25}{100}(11) = \frac{11}{4} = \boxed{2.75}$ → Location

→ $2^{nd} + 0.75(3^{rd} - 2^{nd})$

$= 213 + 0.75(241 - 313)$

$\boxed{Q_1 = 234}$

$Q_2 = \frac{50}{100}(11) = \frac{11}{2} = 5.5$ $\left(\frac{290 + 281}{2}\right)$ (6th, 5th)

$\boxed{Q_2 = 285.5}$

$Q_3 = \frac{75}{100}(11) = 8.25$ → $8^{th} + 0.25(9^{th} - 8$

$= 312 + 0.25(350 + 321)$

$\boxed{Q_3 = 328.25}$

$IQR = Q_3 - Q_1 = 328.25 - 234$

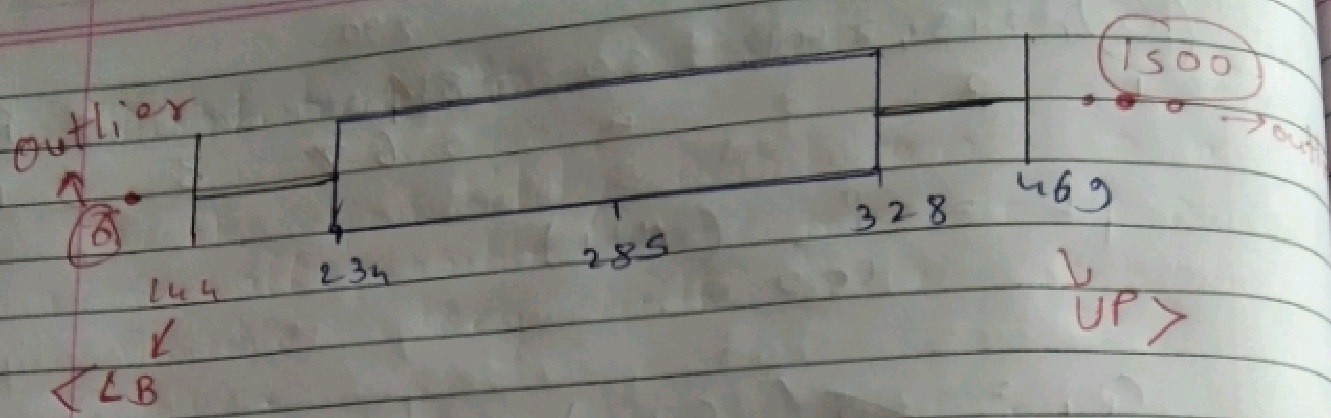
$\boxed{IQR = 94}$

$LB = Q_1 - 1.5(IQR) = 285.5 - 1.5(94)$

$= 144$

$UP = Q_3 + 1.5(IQR) = 328 + 1.5(94)$

$= 469$



---------- X ----------

Probability :- It is the likelyhood or the possibility of event Happening

Tossing a coin = {H, T}

probability of event ~~Happe~~ Head = $\left(\frac{\text{event}}{\text{Total outcome}}\right)$ $\frac{1}{2}$

$= \frac{1}{2} = 0.5$

possibility ← $P(T) = 1 - P(H)$

$= 1 - 0.5$

$= 0.5$

Date.

Types of variable

Algebric Variable

$5 = 10 + x$ $= 5 - 10 = x$

fix value $\Rightarrow$ $\boxed{x = -5}$

Random Variable :- not a fix value

$x = \{\text{Tossing a coin}\}$ $x = \{H, T\}$

$x = \{\text{Throwing a Dice}\}$ $x = \{1, 2, 3, 4, 5, 6$

Random variable

Discrete (Whole no.)

Continious RV (Height, weight

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x

Distribution :-

The probability Distribution is a list of all possible outcomes of a Random Variable along with their probability value.

outcomes	H	T
Probability	1/2	1/2

1 Dice

6	5	4	3	2	1	outcomes
1/6	1/6	1/6	1/6	1/6	1/6	Probability

2 Dice

Dice 1 →

Dice 2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$P(2) = \frac{1}{36}$ $\quad P(4) = \frac{3}{36}$ $\quad P(6) = \frac{5}{36}$

$P(3) = \frac{2}{36}$ $\quad P(5) = \frac{4\cancel{8}}{36}$ $\quad P(7) = \frac{6}{36}$

$P(8) = \frac{5}{36}$

$P(9) = \frac{4}{36}$

$P(10) = \frac{3}{36}$

$P(11) = \frac{2}{36}$

$P(12) = \frac{1}{36}$